

Neonatal Cholestasis: Epidemiology and Definitions.

Picture a well baby at the two-week check. She is feeding at the breast, gaining weight, and her parents mention that she still looks a little yellow. Most babies like her are perfectly healthy. A few of them have a liver that is failing to drain bile, and one of those few may have biliary atresia, a disease in which the bile ducts outside the liver are being destroyed. For that child, the date of surgery decides how long she keeps her own liver. This chapter is about how we find that one child among hundreds of healthy yellow babies.

The clinical question is simple to state. Which jaundiced infant needs a fractionated bilirubin, and what result counts as abnormal? The main idea of the chapter is just as simple. Neonatal cholestasis is defined by a number and found by a habit. The number is a conjugated or direct bilirubin above one milligram per decilitre, whatever the total bilirubin is. The habit is to measure that fraction in every infant who is still jaundiced at two weeks of age, or at three weeks if the baby is breastfed and completely well, and on the same day at any age if the stool is pale.

We will work through this in order. First, the mechanism: why conjugated bilirubin is never a normal finding in a newborn. Second, how common neonatal cholestasis is, and where the familiar figure of one in two thousand five hundred comes from. Third, how often each cause is found. Fourth, how the definition is written in different guidelines, and why the old percentage rule was abandoned. Fifth, the laboratory problem of direct versus conjugated bilirubin. Sixth, when to test, with special attention to the breastfed baby. And finally, why the clock matters so much.

A short note on sources. The main document is the 2017 guideline on cholestatic jaundice in infants from the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, called NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, called ESPGHAN, written by Fawaz and colleagues. We also use the 2022 guideline of the American Academy of Pediatrics, the AAP, on hyperbilirubinaemia in newborns, written by Kemper and colleagues, and the AAP clinical report on early detection of biliary atresia published in 2025 by Harpavat and colleagues. From the United Kingdom we use the guideline on neonatal jaundice from the National Institute for Health and Care Excellence, called NICE, known as clinical guideline ninety-eight. Neither the American Association for the Study of Liver Diseases nor the European Association for the Study of the Liver has a guideline on how to assess the cholestatic infant.

Part one. Why conjugated bilirubin is never physiological.

Bilirubin comes from the breakdown of haem, mostly from old red blood cells. In its first form it is unconjugated. Unconjugated bilirubin does not dissolve in water. It travels in the blood bound to albumin, it cannot pass into urine, and in high amounts it can cross into the brain and cause kernicterus.

The liver takes up unconjugated bilirubin and attaches one or two molecules of glucuronic acid to it, using the enzyme uridine diphosphate glucuronosyltransferase one A one, usually shortened to UGT one A one. The product, conjugated bilirubin, is water-soluble. The hepatocyte then pumps it into the bile canaliculus through a transporter called multidrug resistance-associated protein two, or MRP two, and from there it flows through the bile ducts into the gut.

A newborn has every reason to have unconjugated jaundice. The baby has a large red-cell mass, red cells that live a shorter time, an immature UGT one A one enzyme, and more reabsorption of bilirubin from the gut. So unconjugated jaundice in the first weeks is expected, and when it lasts longer in a breastfed baby it is usually harmless.

Conjugated bilirubin is different. Conjugation works early enough that a healthy newborn has very little conjugated bilirubin in the blood. In a study of more than two hundred and seventy thousand term newborns in Northern California, published by Davis and colleagues in 2011, the ninety-ninth centile of conjugated bilirubin was only half a milligram per decilitre. If conjugated bilirubin builds up in the blood, either the hepatocyte cannot export it or bile cannot leave the liver. There is no stage of normal maturation that explains it. That is why a raised conjugated bilirubin is treated as a disease until proven otherwise.

At the cot side, conjugated bilirubin gives two clues that unconjugated bilirubin never gives. Because it is water-soluble, it darkens the urine. And because bile is not reaching the gut, the stool becomes pale, clay-coloured or white. A useful way to remember all this is that prolonged unconjugated jaundice is usually a feeding question, while conjugated jaundice is a liver question.

Part two. How common is neonatal cholestasis?

The number quoted everywhere, including in the 2017 guideline, is about one in every two thousand five hundred term infants. It is worth knowing where this comes from. It comes from a prospective survey by Dick and Mowat, published in the Archives of Disease in Childhood in 1985. Between 1971 and 1973 they identified every infant in a defined area of south-east England with conjugated jaundice lasting more than two weeks. They found fifty-four infants, about one per two thousand five hundred live births. Twenty-nine of them were labelled as having cryptogenic hepatitis, meaning no cause was found. Eleven had biliary atresia. Seven had alpha-one antitrypsin deficiency. The rest included three congenital infections, one chromosomal disorder, and one case each of choledochal cyst, galactosaemia and rhesus disease.

Two lessons follow. Our best population figure is about fifty years old and comes from one region. And it was gathered before genetic testing, so more than half of those infants were called cryptogenic. Many of them today would receive a genetic diagnosis.

The picture in the neonatal intensive care unit is very different. In a Turkish tertiary unit, conjugated hyperbilirubinaemia was found in two point one percent of admissions, one hundred and four of four thousand nine hundred and fifteen babies, and four out of five of those babies had another illness at the same time. In a United States unit studied by Sloane and colleagues, one point seven percent of infants had an abnormal direct bilirubin within the first seventy-two hours. That is roughly fifty times the rate in the general newborn population.

Parenteral nutrition, the feeding of a baby through a vein, is a major reason. Christensen and colleagues studied six thousand five hundred and forty-three neonates who received parenteral nutrition, and defined cholestasis as a direct bilirubin of two milligrams per decilitre or more. Among babies fed this way for fourteen to twenty-eight days, about fourteen percent became cholestatic. Among those fed this way for more than one hundred days, about eighty-five percent did. The highest risk was in babies born weighing less than five hundred grams, with an odds ratio of about thirty-one,

and in babies with gastroschisis or jejunal atresia, with odds ratios of about twenty and twenty-four. The important trap is that a sick preterm baby on parenteral nutrition can also have biliary atresia. A plausible explanation does not end the work-up.

Part three. How common are the named causes?

Biliary atresia varies a lot between populations. In England and Wales, between 1999 and 2006, it affected about one in seventeen thousand live births, or zero point five eight per ten thousand. In metropolitan France the rate was about zero point five one per ten thousand, and in the United States, from billing data, about zero point four five per ten thousand. In Taiwan, the national registry recorded one point seven nine per ten thousand in 2004 to 2006 and one point two three per ten thousand in 2007 to 2009. In French Polynesia the rate was about two point nine per ten thousand, and in New Zealand it was about one in five thousand three hundred among Maori infants, compared with about one in sixteen thousand among infants of European ancestry. In other words, Pacific and East Asian infants carry a three- to sixfold higher prior probability. The same pale stool means more in those babies. In the United States the recorded incidence rose by almost eight percent a year between 1997 and 2012, and the median age at the Kasai operation was sixty-three days, which is too late.

Alpha-one antitrypsin deficiency is much commoner at birth. In the classic Swedish screening study by Sveger, published in 1976, two hundred thousand newborns were screened and one hundred and twenty had the PiZ form, about one in one thousand six hundred and seventy. But only fourteen of those one hundred and twenty had prolonged obstructive jaundice in infancy. So this disease is about ten times commoner than biliary atresia at birth, yet only about one affected infant in nine becomes jaundiced.

Alagille syndrome has a clinical frequency of about one in seventy thousand to one in one hundred thousand births. When relatives with the gene change but only mild features are counted, as Kamath and colleagues showed in 2003, the true frequency is probably close to one in thirty thousand. Citrin deficiency, which causes a neonatal cholestasis that looks a little like galactosaemia and comes with a high citrulline, is common in East Asia. The carrier rate of the responsible gene, called SLC twenty-five A thirteen, is about one in sixty-nine people in Japan, one in sixty-five in China, about one in forty-eight in southern China, and one in one hundred and twelve in Korea. Ancestry is therefore part of the history. The progressive familial intrahepatic cholestasis diseases are individually rare and made up about one percent of cases in pooled series. Their clue is a normal or low gamma-glutamyl transferase, or GGT, despite cholestasis.

Part four. What does the jaundiced infant turn out to have?

The best summary of causes comes from a systematic review by Gottesman, Del Vecchio and Aronoff, published in BMC Pediatrics in 2015, which pooled one thousand six hundred and ninety-two infants with conjugated hyperbilirubinaemia. Idiopathic neonatal hepatitis accounted for twenty-six percent. Extrahepatic biliary atresia accounted for almost exactly the same share, about twenty-six percent. Infection accounted for about eleven and a half percent, and parenteral nutrition for about six and a half percent. Metabolic diseases were about four and a half percent, alpha-one antitrypsin deficiency about four percent, and perinatal hypoxia or ischaemia between three and four percent. Paucity of the interlobular bile ducts was about two and a half percent, choledochal cyst about

two percent, and hypopituitarism or hypothyroidism about two percent. Haemolysis, inspissated bile, progressive familial intrahepatic cholestasis, Alagille syndrome and cystic fibrosis each made up about one percent, and other causes about six percent, including trisomy twenty-one and cholestasis of prematurity.

So one infant in four has biliary atresia and one in four still has no name for the disease. The treatable causes, such as infection, hypothyroidism, hypopituitarism and galactosaemia, are small slices of the whole. They are small, but they are exactly the ones you must not miss, because each has a specific treatment.

Part five. One number: one milligram per decilitre.

The definition has changed over time. The 2004 NASPGHAN guideline used a direct bilirubin above one milligram per decilitre when the total was low, but switched to a percentage of the total, usually twenty percent, when the total bilirubin was high. The 2017 NASPGHAN and ESPGHAN guideline dropped the percentage. It states that a conjugated or direct bilirubin above one milligram per decilitre, which is seventeen micromoles per litre, is pathological and needs evaluation, and that a specific conjugated measurement is best where it is available. The 2022 AAP guideline says the same thing: a direct bilirubin above one milligram per decilitre is abnormal, and a direct fraction above twenty percent of the total is no longer regarded as necessary. It adds a practical point. More than ninety-nine percent of infants with a single raised value do not have biliary atresia, and repeating the test within a few days to two weeks makes the result much easier to interpret. The 2025 AAP clinical report treats a direct or conjugated value of one milligram per decilitre or more as a reason for urgent consultation with a gastroenterologist, with the goal of a Kasai operation before thirty days of age.

Two documents use other numbers. The NICE guideline in the United Kingdom asks clinicians to follow expert advice when the conjugated bilirubin is above twenty-five micromoles per litre, which is about one and a half milligrams per decilitre. A 2018 review pointed out that this threshold is not based on any referenced data. A Spanish consensus from 2025 still uses either a direct bilirubin above one milligram per decilitre or above twenty percent of the total. So the old rule still circulates. To convert units, divide micromoles per litre by about seventeen.

Why did the percentage rule fail? The answer comes from a study by Harpavat and colleagues in Houston, published in *Pediatrics* in 2011. They looked back at sixty-one infants with biliary atresia. Thirty-four of them had a direct or conjugated bilirubin measured in the first days of life, and every one of those thirty-four values was above the laboratory's normal range. At twenty-four to forty-eight hours, the mean direct bilirubin was one point four milligrams per decilitre in the infants with atresia, compared with zero point one nine in controls. But seventy-nine percent of the infants with atresia had a direct-to-total ratio of zero point two or less. In the first days, the total bilirubin is high for ordinary reasons, so the conjugated share looks small even when the absolute value is clearly abnormal. The twenty percent rule would have missed four out of five of these babies. The obstruction starts before birth. A ratio hides it and an absolute value shows it.

Part six. Direct is not conjugated.

Many laboratories report direct bilirubin, not conjugated bilirubin, and the two are not the same thing. The direct test uses a diazo reaction. It measures the conjugated forms, meaning the mono- and

diglucuronides, and also a third form called delta bilirubin. Delta bilirubin is bilirubin that has become permanently, covalently bound to albumin. Because it is stuck to albumin, it stays in the blood as long as albumin does, with a half-life of about twenty-one days. A specific conjugated assay measures only the glucuronides.

This matters in three ways. First, the normal range is different. In the Northern California study, the ninety-ninth centile in newborns was zero point five milligrams per decilitre for conjugated bilirubin but two point one milligrams per decilitre for direct bilirubin. The authors concluded that the two are not interchangeable. The AAP guideline makes the same point: direct values are higher and more variable than conjugated values. A direct bilirubin of one point two on day two may be within the newborn range for that assay, while a conjugated bilirubin of one point two is clearly abnormal. Second, delta bilirubin falls slowly. After a successful Kasai operation, or after any cholestasis resolves, the direct bilirubin can stay raised for weeks because of delta bilirubin, even though bile is flowing again. The conjugated value and the stool colour are better guides. Third, a high value really does mean disease. In the same study, when conjugated bilirubin reached five milligrams per decilitre, forty-seven percent of infants had biliary disease and forty-three percent had liver disease.

Two more points from the AAP guideline. When deciding about phototherapy, do not subtract the direct fraction from the total bilirubin. And if the direct fraction is more than half of the total, discuss the baby with a specialist. The practical lesson is to read the laboratory report and know which test your laboratory uses.

Part seven. When to test, and the breastfed baby.

The 2017 guideline says that a formula-fed infant who is still jaundiced after two weeks should be evaluated with a total and a conjugated or direct bilirubin. A breastfed infant who is otherwise completely well, with normal stool and urine colour and a normal examination, may be followed clinically until three weeks of age. The 2022 AAP guideline, in its ninth key action statement, says that breastfed infants still jaundiced at three to four weeks, and formula-fed infants still jaundiced at two weeks, should have the total and direct or conjugated bilirubin measured. The 2025 AAP clinical report describes a three-step check at the two- to four-week visit: look for jaundice beyond two to three weeks, ask about pale stools at any time, and review any earlier direct or conjugated bilirubin results. The NICE guideline defines prolonged jaundice as jaundice lasting more than fourteen days in a baby born at term, or more than twenty-one days in a baby born before thirty-seven weeks. It asks for a conjugated bilirubin together with a look at stool and urine colour, a full blood count, blood group and direct antiglobulin test, consideration of a urine culture, and confirmation that the metabolic and thyroid screening has been done.

Why not simply test the babies who look unwell or very yellow? Because prolonged jaundice in breastfed babies is extremely common. Maisels and colleagues followed one thousand and forty-four mainly breastfed infants born at thirty-five weeks or later, measuring bilirubin through the skin. At about three weeks of age, forty-three percent had a skin bilirubin of five milligrams per decilitre or more and thirty-four percent were visibly jaundiced. At about four weeks, thirty-four percent still had a skin bilirubin of five or more and twenty-one percent were visibly jaundiced. The authors concluded that it is normal for twenty to thirty percent of breastfed infants to be jaundiced at three to four weeks. The eye cannot tell which of these babies has a raised conjugated fraction. Only a blood test can.

Part eight. Why the clock matters.

All of this attention to definitions and timing exists for one reason: biliary atresia gets worse every week, and the Kasai operation works better when it is done early. In a Canadian study of three hundred and forty-nine infants, published by Schreiber and colleagues in 2007, survival with the native liver at four years was forty-nine percent when the Kasai was done at thirty days or less, thirty-six percent when it was done between thirty-one and ninety days, and twenty-three percent when it was done after ninety days. In a French national study of seven hundred and forty-three infants, published by Serinet and colleagues in 2009, native-liver survival fell step by step with each fifteen-day band of age at surgery, and the authors estimated that operating on every child before day forty-six could spare about six percent of paediatric liver transplants each year. A large Japanese series of almost four thousand infants, reported in 2025, showed that about half of the children still had their own liver at ten years and about forty-one percent at thirty years, so a good Kasai can last for decades.

Screening can move the clock. In Texas, Harpavat and colleagues measured direct or conjugated bilirubin in the first days of life in one hundred and twenty-four thousand three hundred and eighty-five newborns at fourteen hospitals between 2015 and 2018, and retested at about two weeks those whose first value was above the cut-off. The screen found all seven infants with biliary atresia, a sensitivity of one hundred percent, although with so few cases the ninety-five percent confidence interval runs from about fifty-six to one hundred percent. The specificity was ninety-nine point nine percent. Because atresia is rare, the positive predictive value was only about six percent, so most screen-positive babies did not have it. The mean age at Kasai fell from fifty-six to thirty-six days. Without screening, practice remains slow. A survey by the ESPGHAN quality of care task force, covering seven hundred and eighty-five infants with atresia in fifteen countries between 2015 and 2019, found a median age of fifty-three days at referral and sixty days at Kasai, and six percent of the infants were already too cirrhotic for the operation. Screening and stool colour cards are the subject of the next chapter. The point for today is that the same logic, an absolute conjugated value above the normal range, works even on the second day of life.

Part nine. Where the guidelines agree and where they differ.

The North American and European paediatric documents agree on most things. They agree that a formula-fed baby who is jaundiced at two weeks should have a fractionated bilirubin. They agree that a conjugated assay is preferred. And they agree that a pale stool at any age should trigger testing. No guideline yet recommends universal newborn screening, and the United Kingdom National Screening Committee does not recommend it, although it opened a new consultation in 2026.

They differ in a few places. The threshold differs, because NICE uses twenty-five micromoles per litre rather than seventeen, and the Spanish consensus keeps the twenty percent rule that the North American guidelines dropped. The most important difference is timing in the breastfed baby. NICE asks for testing at fourteen days for all term babies, the 2017 guideline allows a well breastfed baby to be followed until three weeks, and the 2022 AAP guideline allows up to three to four weeks. A four-week window can push the Kasai operation past thirty days, which is why the 2025 AAP report pulls the check earlier. There are also gaps. The North American documents have no separate timing rule for preterm babies, and the adult liver societies have no guidance on first assessment of the

cholestatic infant. The 2024 EASL guideline on genetic cholestatic liver disease covers the genetic diseases themselves, not the initial approach.

Part ten. What to do differently in clinic.

Here are the habits to take away. Fractionate the bilirubin in every infant who is still jaundiced at the two-week visit, and do not wait until four weeks in a breastfed baby. If the stool is pale, test on the same day at any age. Use an absolute threshold of one milligram per decilitre, not a percentage of the total. Know whether your laboratory reports direct or conjugated bilirubin, remember that the newborn normal range for direct bilirubin is higher, and repeat a borderline direct value as a conjugated value within days to two weeks. After a Kasai operation, judge progress by conjugated bilirubin and stool colour, because delta bilirubin makes the direct value fall slowly. In the neonatal unit, do not let parenteral nutrition explain away a high conjugated bilirubin until biliary atresia has been excluded. And take ancestry into account: an East Asian or Pacific infant has a higher chance of biliary atresia, and an East Asian infant may have citrin deficiency.

Questions to test yourself.

Question one. Where does the figure of one in two thousand five hundred come from? Answer: from the prospective survey by Dick and Mowat of conjugated jaundice lasting more than two weeks in south-east England between 1971 and 1973, which found fifty-four infants.

Question two. What threshold do the 2017 NASPGHAN and ESPGHAN guideline and the 2022 AAP guideline use to define a pathological conjugated or direct bilirubin? Answer: above one milligram per decilitre, or seventeen micromoles per litre, regardless of the total bilirubin.

Question three. Why was the twenty percent rule abandoned? Answer: because in the first days of life the total bilirubin is high, so the ratio looks normal. In the Houston study, seventy-nine percent of infants with biliary atresia had a direct-to-total ratio of zero point two or less, even though their absolute values were all abnormal.

Question four. What are the newborn ninety-ninth centiles for conjugated and direct bilirubin? Answer: about zero point five milligrams per decilitre for conjugated bilirubin and about two point one for direct bilirubin.

Question five. Why can the direct bilirubin stay high after bile flow has been restored? Answer: because it includes delta bilirubin, which is bound to albumin and has a half-life of about twenty-one days.

Question six. What proportion of breastfed infants are visibly jaundiced at three weeks of age? Answer: about a third, and about one in five at four weeks.

Question seven. What were the two largest causes in the Gottesman systematic review? Answer: idiopathic neonatal hepatitis and biliary atresia, each about twenty-six percent.

Question eight. In the Canadian study, how did four-year survival with the native liver change with age at Kasai? Answer: it was forty-nine percent at thirty days or less, thirty-six percent at thirty-one to ninety days, and twenty-three percent after ninety days.

End of chapter.